

ABDURRAHIM ISLAM

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SUMMARY

Computer Science and Information Technology & Web Science (ITWS) student at Rensselaer Polytechnic Institute with hands-on experience in IT infrastructure, systems administration, offensive and defensive cybersecurity, and data analytics. Experience spans manufacturing data-collection and AWS cloud-architecture consulting for a research center, UX/dashboard design for an AI manufacturing platform, and an independently built enterprise-style NAS/Homelab supporting lab environments in Active Directory, SIEM/log analysis (Wazuh, Splunk), penetration testing (Kali Linux, Metasploit, OWASP ZAP), network security, and domain/reverse-proxy architecture. Also a part-time indie animator and game developer using Unreal Engine 5.

EDUCATION

Rensselaer Polytechnic Institute

Grad 2029 - Troy, NY

Bachelor of Science, Computer Science and Information Technology & Web Science (ITWS)

The Albany Academy

Grad 2025 - Albany, NY

High School (GED)

Relevant Coursework: Foundations of Computer Science, Computer Architecture and Operating Systems, Data Structures, Web Systems Development

CERTIFICATIONS

- Google IT Support Certificate - System Administration, Google / Coursera
- CompTIA Security+ (SY0-701) - In Progress
- AWS Certified Solutions Architect - Associate (SAA-C03) - In Progress

TECHNICAL SKILLS

Cybersecurity & Monitoring: Wazuh (SIEM), Splunk (SIEM), log analysis, alert triage, endpoint monitoring, malware scanning/quarantine workflows, access control, risk assessment

Offensive Security: Kali Linux, Metasploit Framework, OWASP ZAP, penetration-testing methodology, vulnerability assessment, web-application security testing

Systems & Identity: Active Directory, Windows Server, Ubuntu Linux, Unraid, Docker, virtual machines, SMB, NAS administration

Networking: Cisco Packet Tracer, Wireshark, TCP/IP, IPv4, DNS, VLANs, subnetting, routing & switching, VPNs, Tailscale, port mapping, network troubleshooting

Cloud & Data Architecture: AWS (S3, EC2, IAM fundamentals), Azure, Cloudflare, reverse-proxy architecture (Nginx Proxy Manager), domain/DNS management, object storage / data-lake design, cloud-vs-on-prem cost and architecture tradeoff analysis

Automation & Integration: Python/Bash scripting, REST API integration, workflow automation (n8n), scheduled task orchestration

Data & Programming: Python, Java, C, C++, R, SQL, MariaDB, PostgreSQL, MongoDB, data visualization, time-series analysis

AI Infrastructure: Local LLM deployment (Ollama), GPU-accelerated inference, private search integration

Web & Design: HTML, CSS, JavaScript, Balsamiq, UI/UX research, responsive design

Soft Skills: Team collaboration, verbal and written communication, presenting technical findings to non-technical stakeholders, troubleshooting & root-cause analysis, attention to detail, self-directed learning

EXPERIENCE

Consultant - Data Analytics & Digital Media

June 2026 - August 2026

(Independent Contractor) | CSCMS — an RPI Manufacturing Research Center | Albany, NY (Hybrid)

- Augmented manufacturing data-collection workflows across 4 machines and a dataset of thousands of operation-critical files, improving structure and traceability for downstream analytics and AI/process-intelligence use.
- Advised on cloud storage architecture, comparing AWS S3 against document-oriented storage for machine-generated manufacturing data; modeled a 260+ TB illustrative production scenario (video, thermal imaging, sensor logs) to frame object-storage and AI/ML pipeline requirements.
- Evaluated local-hosted infrastructure vs. SaaS tradeoffs across cost, control, and scalability, and coordinated an internal network-switch effort to support the assessment.

SMRCP - UI/UX Systems Integration

June 2026 - August 2026

Front-End & Product Design Engineer / UX Research Lead, Smart Manufacturing Research Co-Pilot | Albany, NY (Remote)

- Designed and prototyped interfaces in Figma for an edge-based AI platform enabling natural-language interaction with cyber-physical manufacturing systems.
- Built dashboards visualizing real-time and historical multi-modal sensor data, applying time-series analysis and sensor-fusion concepts.
- Conducted user interviews and focus groups and translated findings into product and interface improvements.

Associate / Social Media Manager, Universal Concepts ITP

June 2020 - Present

Albany, NY

- Design and maintain the website and digital media presence for a 501(c)(3) youth workforce-development nonprofit.

TECHNICAL PROJECTS — INFRASTRUCTURE & SECURITY LABS

Homelab - NAS Storage & Infrastructure Architecture (Unraid)

- Self-built a Network-Attached Storage (NAS) system from scratch using a mix of spare PC parts and enterprise-grade storage hardware.

- Rebuilt the storage array around a flashed IT-mode HBA and four 4 TB SAS drives; the HDDs form a parity-protected array using the XFS file system, totaling ~20 TB raw across parity and data disks (~16 TB usable).
- Built a mixed NVMe/SSD cache pool with a BTRFS file system from a 500 GB SSD and a 1 TB M.2 NVMe (~750 GB usable) and a dedicated 250 GB quarantine SSD.
- Operated Docker containers to host all application and VM workloads.
- Deployed a dedicated network switch to control device access and an uninterruptible power supply (UPS) for power-outage protection.
- Recovered full Docker/VM application state after a cache-format incident, rebuilding with clearer separation between application data, production data, and caches.

Windows Server & Active Directory Domain Lab

- Deployed a Windows Server Active Directory environment in a VM, creating and managing user and group objects and Organizational Units (OUs), and validated end-to-end domain-user authentication from a client system.
- Configured Group Policy Objects (GPOs) to enforce domain-wide password policy (complexity, expiration, account lockout) and performed user password resets.
- Independently diagnosed and resolved VM console, networking, and remote-management issues.

Offensive Security Lab (Kali Linux)

- Deployed Kali Linux as a dedicated offensive-security workstation and practiced reconnaissance, enumeration, and vulnerability-assessment workflows against a self-owned, isolated VM lab.
- Built practical familiarity with Linux-based penetration-testing tooling and attacker-perspective methodology, complementing defensive monitoring work in the Wazuh lab.

Wazuh Security Monitoring (SIEM) Lab

- Built a Wazuh environment for centralized log collection, agent-based endpoint telemetry, and alert triage, mapping detections to the MITRE ATT&CK framework and tuning correlation rules to reduce false positives.

Splunk SIEM Lab

- Practiced SIEM/log-analysis workflows in Splunk as a second platform alongside Wazuh, correlating log data and reinforcing detection-rule and indicator-of-compromise (IOC) concepts across different tooling.

Network Traffic Analysis (Wireshark)

- Captured and analyzed live traffic on a personal home network using Wireshark, applying display filters for protocol-level inspection and developing network-visibility skills that reinforced later networking and VPN work.

Exploitation Framework Practice (Metasploit)

- Practiced exploitation workflows using the Metasploit Framework, including exploit modules and payloads, against controlled lab targets in the isolated VM environment, working the full path from vulnerability identification to exploit execution and validation.

Cisco Packet Tracer Enterprise Networking Lab

- Modeled enterprise network topologies and practiced IPv4 addressing, routing/switching, and device configuration, using structured troubleshooting to isolate and resolve connectivity errors.

Web Application Security Lab (OWASP ZAP + Juice Shop)

- Deployed OWASP Juice Shop as a deliberately vulnerable target and used OWASP ZAP as an intercepting proxy to assess OWASP Top 10 web-application vulnerabilities from an attacker's perspective.

Domain, DNS & Reverse-Proxy Architecture

- Purchased and configured a personal domain and built a Cloudflare/Nginx Proxy Manager reverse-proxy layer to publish select self-hosted services under a controlled, authenticated namespace.
- Maintained strict LAN/Tailscale-only access for sensitive administrative infrastructure, treating the forward-facing boundary as a deliberate architectural decision rather than open internet exposure.

Ubuntu Linux Administration

- Administered Ubuntu Linux systems supporting security tooling, AI infrastructure, and automation workloads, managing users, permissions, packages, and networking via the command line.

Aflix - Personal Streaming Service (Jellyfin)

- Built and operate a personal media streaming service, Aflix, running on a self-hosted Jellyfin server, including automated video transcoding to optimize playback across devices and network conditions.
- Wrote an automated media-importation pipeline that routes every incoming file through a scheduled malware scan before it reaches the library, replacing a manual process with a scan-then-approve-then-import workflow.
- Validated the import pipeline end-to-end with a 37-item batch, achieving 37/37 successful, correctly identified imports after resolving a file-parsing issue.
- Diagnosed and corrected a scan-logic edge case that could misclassify an indeterminate result as a positive detection.

AI Infrastructure Assistant Architecture (n8n + Ollama) — In Progress

- Designing a staged, safety-conscious architecture for a local AI infrastructure assistant using n8n and Ollama, incorporating read-only tooling, allowlisted commands, and mandatory human confirmation before any state-changing action; Phase 1 implementation is in progress.

Local AI Infrastructure

- Deployed a GPU-accelerated local AI stack (LLM serving, web interface, private search) on a 12 GB RTX 4070 GPU; evaluated model sizes from 7B up to 50B+ parameters against VRAM/offload tradeoffs.

Independent Animation & Game Development (Unreal Engine 5)

- Build interactive 3D scenes, animations, and short-form game experiences independently using Unreal Engine 5 and Blender, working across environment design, asset creation, and character animation.

ACTIVITIES

- National Society of Black Engineers (NSBE), African Students Association (ASA), Boxing, Rugby